10.3 Transport on Track (TRA)

A study of a modern rail transport system with an emphasis on safety and control:

- track circuits and signalling
- sensing speed
- mechanical braking
- regenerative and eddy current braking
- crash-proofing.

Students use mathematical models to describe the behaviour of moving vehicles and to model electromagnetic induction and capacitor discharge.

There are opportunities to develop information and communication technology skills.

There are opportunities for students to discuss ethical, environmental and other issues relating to decisions about transport taken by government, by transport companies and by individuals.

Students will be assessed on their ability to:	Suggested experiments
73 use the expression $p = mv$	
74 investigate and apply the principle of conservation of linear momentum to problems in one dimension	Use of, for example, light gates and air track to investigate momentum
75 investigate and relate net force to rate of change of momentum in situations where mass is constant (Newton's second law of motion)	Use of, for example, light gates and air track to investigate change in momentum
78 explain and apply the principle of conservation of energy, and determine whether a collision is elastic or inelastic	
87 investigate and use the expression $C = Q/V$	

Students will be assessed on their ability to:	Suggested experiments
89 investigate and recall that the growth and decay curves for resistor–capacitor circuits are exponential, and know the significance of the time constant RC	
90 recognise and use the expression $Q = Q_0 e^{-t/RC}$ and derive and use related expressions, for exponential discharge in RC circuits, for example, $I = I_o e^{-t/RC}$	Use of data logger to obtain I-t graph
91 explore and use the terms magnetic flux density B, flux Φ and flux linkage $N\Phi$	
92 investigate, recognise and use the expression $F = BIl \sin \theta$ and apply Fleming's left hand rule to currents	Electronic balance to measure effect of I and l on force
94 investigate and explain qualitatively the factors affecting the emf induced in a coil when there is relative motion between the coil and a permanent magnet and when there is a change of current in a primary coil linked with it	Use a data logger to plot V against t as a magnet falls through a coil of wire
95 investigate, recognise and use the expression $\varepsilon = d(N\Phi)/dt$ and explain how it is a consequence of Faraday's and Lenz's laws	